MAJOR PROJECT REPORT

ON

SMART STREET MONITORING SYSTEM USING ULTRASONIC AND LDR SENSORS

By

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Submitted to

DEPARTMENT OF COMPUTER APPLICATIONS

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BONAFIDE CERTIFICATE

This is to certify that the Major Project entitled **"SMART STREET MONITORING SYSTEM USING ULTRASONIC AND LDR SENSORS"** is the Bonafide work of **HARI PRASATH A (Reg No: RA2432241050062)**, who completed the project under my supervision at SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, Tiruchirappalli, in the year 2026.

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HARI PRASATH A

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ABSTRACT

The increasing demand for energy-efficient and intelligent lighting systems has led to the development of smart street lighting solutions. This project presents a Smart Street Light Monitoring System that utilizes an ultrasonic sensor and a Light Dependent Resistor (LDR) to optimize power consumption and improve automation in urban and rural lighting infrastructure.The system is designed to operate based on environmental lighting conditions and object detection. The LDR sensor serves as the primary ambient light detector, distinguishing between daytime and nighttime conditions. During daytime, when the LDR detects adequate ambient light, the entire system remains inactive to conserve electrical energy. During nighttime, when ambient light falls below the threshold level, the ultrasonic sensor is activated to continuously monitor the distance of approaching objects, including vehicles, pedestrians, cyclists, and other moving entities.Based on the detected distance, the system dynamically controls two distinct levels of street lighting using relay modules connected to the Arduino UNO microcontroller. When an object is detected within the range of 10 centimeters, a low-intensity light operating at 7V is activated to provide basic illumination. When the object approaches closer, within 5 centimeters, a high-intensity light operating at 12V is triggered to ensure maximum visibility and road safety. In the absence of any detected object beyond 10 centimeters, the street lights remain completely turned off, thus conserving energy during periods of low traffic.This intelligent multi-level switching mechanism significantly reduces power consumption compared to conventional street lighting systems, in which lights remain continuously operational throughout the night regardless of traffic conditions. The proposed system is cost-effective, easy to implement using commercially available components, and suitable for real-time applications in smart cities, highways, expressways, and rural areas where consistent electricity supply may be limited.

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid advancement of technology in the twenty-first century has fundamentally transformed the way human societies manage and consume energy. Among the numerous sectors that consume substantial amounts of electrical energy, public street lighting constitutes one of the most significant contributors to urban electricity expenditure. Across the globe, millions of street lights operate throughout the night, consuming vast quantities of electrical power irrespective of whether roads are occupied by traffic or completely deserted. This indiscriminate use of energy represents not only a massive financial burden on municipal authorities and governments but also a significant contributor to environmental pollution and carbon emissions.

Conventional street lighting systems, predominantly based on sodium vapor lamps, mercury vapor lamps, or fluorescent tubes, operate on a simplistic ON/OFF mechanism that is triggered either by a fixed timer or by a basic ambient light sensor. While these systems mark a degree of improvement over fully manual systems, they are fundamentally incapable of adapting to real-time traffic conditions. A street light controlled solely by a timer or a simple LDR continues to burn at full intensity throughout the entire night, regardless of whether a vehicle or pedestrian is present in its vicinity. The result is enormous wastage of electrical energy during periods of minimal or zero traffic, which typically constitutes the majority of nighttime hours in most urban and semi-urban locations.

The global energy crisis, coupled with the escalating concerns about climate change and the depletion of non-renewable energy resources, has created an urgent imperative for the development and deployment of energy-efficient technologies in every sector. Governments worldwide have adopted ambitious targets for reducing energy consumption and carbon emissions, and smart infrastructure has been identified as a key enabler in achieving these targets. The concept of smart cities, which emphasizes the integration of digital technologies and intelligent systems into urban infrastructure, has gained considerable momentum in recent years.

In this context, smart street lighting systems have emerged as one of the most promising areas of innovation. By incorporating sensors, microcontrollers, and communication technologies, these systems can dynamically adapt to real-time environmental and traffic conditions, turning lights on or off or adjusting their intensity based on actual need. This adaptive capability can result in energy savings of up to 50 to 80 percent compared to traditional fixed-schedule lighting systems, representing a transformative improvement in the efficiency of public lighting infrastructure.

The development of low-cost, high-performance embedded hardware platforms, such as the Arduino family of microcontrollers, has democratized the process of building intelligent sensor-based systems. Combined with affordable and reliable sensors such as the HC-SR04 ultrasonic distance sensor and the Light Dependent Resistor (LDR), it is now possible to construct fully functional smart street light systems at a fraction of the cost of commercial solutions. This accessibility makes sensor-based smart lighting a viable option not only for well-funded urban municipalities but also for resource-constrained rural communities and developing nations.

The present project is motivated by these considerations and aims to design, build, and test a Smart Street Light Monitoring System that utilizes an ultrasonic sensor and an LDR to achieve intelligent, adaptive, and energy-efficient street lighting. The system embodies the principles of embedded systems engineering, sensor fusion, and real-time decision-making, making it a valuable contribution to the field of smart infrastructure.

1.2 Problem Statement

The central problem addressed by this project is the chronic energy inefficiency of conventional street lighting systems. Despite accounting for approximately 19 percent of global electricity consumption and 6 percent of greenhouse gas emissions, the lighting sector has been slow to adopt adaptive technologies in the domain of public street lighting. The following specific problems have been identified:

- Continuous Operation at Full Power: Conventional street lights operate at full intensity from dusk to dawn, regardless of traffic density. During late-night and early- morning hours when traffic is minimal, this results in enormous and unnecessary energy consumption.
- Lack of Adaptive Control: Existing systems controlled by timers operate on fixed schedules that cannot account for seasonal variations in daylight hours, unexpected weather conditions, or real-time traffic patterns.
- High Maintenance and Operational Costs: The continuous operation of conventional lighting systems results in accelerated lamp degradation, requiring frequent replacements and incurring high maintenance costs for municipal authorities.
- Environmental Impact: The excessive electricity consumption of conventional street lights, particularly those powered by fossil fuel-based grids, contributes significantly to carbon dioxide emissions and other environmental pollutants.
- Inadequate Safety in Dynamic Traffic Conditions: Fixed-intensity street lights do not adapt to the presence or absence of pedestrians and vehicles, potentially providing inadequate illumination when most needed and wasting energy when illumination is unnecessary.
- Limited Applicability in Remote Areas: In rural and remote areas with limited electricity supply, the high energy demands of conventional street lighting make adequate public illumination economically unfeasible.

These problems collectively highlight the urgent need for a smart, sensor-driven street lighting solution that can dynamically adapt to real-time conditions, thereby reducing energy consumption, lowering operational costs, improving road safety, and contributing to environmental sustainability.

1.3 Objectives

The primary and secondary objectives of this project are as follows:

1.3.1 Primary Objectives

To design and implement a Smart Street Light Monitoring System that automatically controls street light intensity based on real-time object detection and ambient light conditions.To utilize the HC-SR04 ultrasonic sensor for accurate and reliable distance measurement of approaching objects such as vehicles and pedestrians.To employ an LDR sensor for automatic day/night detection, ensuring that the system remains inactive during daytime to conserve energy.To implement a two-level lighting control mechanism using relay modules, providing low-intensity illumination for objects detected at a moderate distance and high- intensity illumination for objects in close proximity.To program the Arduino UNO microcontroller with efficient embedded C code that implements the decision logic for sensor-based lighting control.

1.3.2 Secondary Objectives

To demonstrate significant reduction in energy consumption compared to conventional fixed-intensity street lighting systems through experimental measurement and analysis.To develop a cost-effective prototype using commercially available, off-the-shelf components suitable for mass deployment.To establish a scalable and modular system architecture that can be extended to incorporate additional sensors, wireless communication, and IoT connectivity in future iterations. To contribute to the body of knowledge in the field of embedded systems, smart infrastructure, and energy-efficient lighting technology. To provide a well-documented technical reference for future research and development in smart street lighting systems.

1.4 Scope of the Project

The scope of this project encompasses the following aspects:

Hardware Design: The design and physical assembly of the smart street light prototype, including the selection, procurement, and integration of all hardware components.Embedded Software Development: The development of Arduino firmware in embedded C that implements the sensor reading, distance calculation, LDR threshold detection, and relay control logic.System Testing and Validation: Experimental testing of the assembled prototype under controlled conditions to verify correct functionality, measure sensor accuracy, and assess energy savings.Documentation: Comprehensive technical documentation of the system design, implementation, and results in accordance with IEEE academic standards.The scope explicitly does not include wireless communication modules, IoT cloud integration, solar power harvesting, or multi-node network architectures. These extensions are identified as directions for future work. The system is designed and tested as a standalone, embedded prototype suitable for demonstrating the fundamental principles of sensor-driven adaptive street lighting.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Existing Street Lighting Systems

The evolution of street lighting technology can be traced through several distinct phases, each characterized by advances in lamp technology, control mechanisms, and efficiency. An understanding of existing systems provides the essential context for appreciating the innovations introduced by smart sensor-based approaches.

2.1.1 Conventional Incandescent and Fluorescent Street Lights

The earliest electric street lights utilized incandescent lamps, which are characterized by low luminous efficacy (approximately 10 to 17 lumens per watt), extremely short operational lifespans, and high heat generation. While these lamps provided adequate illumination, their high energy consumption and frequent need for replacement made them economically and environmentally unsustainable for large-scale deployment. Fluorescent lamps, introduced subsequently, offered improved efficacy (50 to 100 lumens per watt) and longer lifespans, but still operated on fixed schedules controlled by manual switches or basic timers.

2.1.2 High-Pressure Sodium (HPS) and Metal Halide Lamps

High-Pressure Sodium (HPS) lamps became the dominant technology for street lighting during the latter half of the twentieth century, primarily due to their superior luminous efficacy (80 to 150 lumens per watt) and long operational lifespans (up to 24,000 hours). Metal halide lamps, offering better color rendering, were also widely adopted. However, both technologies are characterized by delayed warm-up times, inability to dim efficiently, and high mercury content, raising environmental concerns regarding disposal. The control mechanisms for these lamp types remained fundamentally unchanged—fixed timer-based or basic photocell-based switching—with no provision for adaptive intensity control based on real-time conditions.

2.1.3 LED-Based Street Lighting Systems

Light Emitting Diode (LED) technology has revolutionized the street lighting industry in the twenty-first century. LED street lights offer luminous efficacies exceeding 150 lumens per watt in modern implementations, operational lifespans of 50,000 to 100,000 hours, instant-on capability without warm-up delays, and excellent dimming characteristics. Furthermore, LEDs contain no mercury and are environmentally significantly cleaner than their predecessors. The adoption of LED street lights has enabled the development of dimming- capable systems that can adjust light output in response to external control signals, laying the technical groundwork for smart adaptive lighting.

2.1.4 Timer-Controlled Systems

Timer-controlled street lighting systems represent an early attempt at energy management by programming specific on and off times. While these systems offer improvement over fully continuous operation, they suffer from inherent inflexibility, as they cannot account for variations in sunrise and sunset times across seasons, unexpected overcast conditions during daytime, or real-time traffic patterns during nighttime. Scheduled systems may leave lights burning at full intensity during late-night periods of zero traffic or may fail to provide adequate illumination during unusually dark daytime conditions.

2.1.5 Photocell-Based Automatic Switching

Photocell or photoresistor-based systems represent a more adaptive approach to street light control, automatically switching lights on when ambient light levels fall below a predetermined threshold and off when ambient light levels rise above the threshold. This mechanism eliminates the need for fixed timers and automatically adjusts to seasonal variations in daylight hours. However, these systems remain fundamentally binary in their response—lights are either fully on or fully off—and provide no mechanism for adaptive intensity control based on traffic conditions.

2.2 Related Works and Research Review

A substantial body of research has been conducted in the area of smart and adaptive street lighting systems. The following subsections present a comprehensive review of significant related works from the academic literature.

2.2.1 PIR Sensor-Based Adaptive Lighting

Numerous research groups have explored the use of Passive Infrared (PIR) sensors for detecting the presence of humans and vehicles in the vicinity of street lights. PIR sensors detect changes in infrared radiation caused by the movement of warm bodies through their field of view. Systems based on PIR sensors activate street lights only when human or vehicular presence is detected, returning to a low-power standby state or completely turning off when no presence is detected. The primary limitation of PIR-based systems is their sensitivity to ambient temperature fluctuations, which can cause false triggers, and their inability to provide quantitative distance information, which would enable proportional intensity control.

2.2.2 Microcontroller-Based Intelligent Street Lighting

Research published in the IEEE Transactions on Smart Grid and related journals has demonstrated the feasibility of microcontroller-based intelligent street lighting systems. These systems employ microcontrollers such as the Arduino, PIC, or ARM Cortex series to implement sophisticated control algorithms that process sensor data, make lighting decisions, and manage relay or PWM-based intensity control. The integration of microcontrollers enables far more complex and adaptive decision logic compared to simple transistor-based switching circuits, including the implementation of multi-threshold distance-based intensity control as proposed in the present work.

2.2.3 Ultrasonic Sensor Applications in Traffic and Proximity Detection

The HC-SR04 ultrasonic sensor and its variants have been extensively studied and deployed in proximity detection applications. Research has demonstrated that ultrasonic sensors are highly effective for vehicle detection in road environments due to their ability to detect objects regardless of color, transparency, or surface texture—limitations that affect optical sensors. Studies have also confirmed the reliability of ultrasonic sensors under varying weather conditions, although performance degradation has been noted in heavy rain or fog due to interference with the ultrasonic beam. The integration of ultrasonic sensors with microcontrollers for distance-proportional lighting control has been demonstrated in prototype systems, confirming the technical approach employed in the present project.

2.2.4 LDR-Based Ambient Light Detection Systems

Light Dependent Resistors have been widely used in ambient light sensing applications for several decades. Their simplicity, low cost, and reliability make them an attractive choice for day/night detection in low-cost embedded systems. Research has demonstrated that LDR- based systems, when properly calibrated with appropriate threshold voltage divider circuits, can reliably distinguish between daytime and nighttime conditions with response times in the range of milliseconds. The combination of LDR-based day/night detection with active sensor- based night monitoring represents a dual-layer approach to energy management that is both effective and economical.

2.2.5 GSM and IoT-Based Remote Monitoring Systems

A significant thread of research in smart street lighting has focused on the integration of GSM modules, WiFi, Zigbee, LoRa, or other wireless communication technologies to enable remote monitoring and control of street light networks. These systems enable central control room operators to remotely monitor the status of individual street lights, detect faults or lamp failures in real time, adjust lighting schedules or intensity levels, and generate automated maintenance alerts. While such systems offer significant operational advantages, they also introduce additional hardware complexity, communication infrastructure requirements, and cybersecurity considerations. The present project does not incorporate wireless communication but acknowledges it as an important direction for future enhancement.

2.2.6 Solar-Powered Smart Street Lighting

The combination of smart sensor-based control with solar power harvesting has been explored as a means of creating self-sufficient, off-grid street lighting solutions. These systems utilize photovoltaic panels to charge battery banks during daytime, which then power LED street lights equipped with smart control systems during nighttime. The adaptive, sensor-driven nature of smart lighting control is particularly valuable in solar-powered systems, where battery energy is a limited resource that must be managed carefully to ensure adequate illumination throughout the night. Research has demonstrated that adaptive control can extend battery life by 40 to 60 percent compared to fixed-intensity solar street lights, making smart control a critical enabling technology for solar-powered off-grid street lighting.

2.2.7 Energy Harvesting and Wireless Sensor Networks

Advanced research has explored the deployment of wireless sensor networks for coordinated, city-scale smart street lighting management. In these architectures, individual street light nodes communicate with neighboring nodes and with a central management system, enabling coordinated dimming strategies, predictive fault detection, and centralized energy management. Such systems represent the leading edge of smart street lighting technology and are being piloted in several smart city initiatives worldwide. However, their high implementation complexity and cost currently limit their applicability to large-scale urban deployments with substantial capital investment.

2.3 Comparative Analysis of Reviewed Systems

The following table presents a comparative analysis of the reviewed smart street lighting systems with respect to key performance and implementation parameters.

Table 1: Comparison of Existing Street Light Systems

System Type	Sensor Used	Adaptive Control	Energy Savings	Cost	Complexity
Conventional Fixed	None	No	0%	Low	Very Low
Timer-Based	None	Partial	10–15%	Low	Low
Photocell-Based	LDR	Day/Night Only	20–30%	Low	Low
PIR-Based	PIR	Presence Detection	30–50%	Medium	Medium

Ultrasonic + LDR	HC-SR04, LDR	Distance Adaptive	50–70%	Medium	Medium
IoT / GSM Enabled	Multiple	Full Remote Control	60–80%	High	High
Solar + Smart	PV + Multiple	Complete Automation	70–90%	Very High	Very High

The comparative analysis reveals that the proposed ultrasonic and LDR-based smart street lighting system occupies an optimal position in the design space, offering substantial energy savings (estimated 50 to 70 percent) at a moderate implementation cost and complexity level. It surpasses conventional and photocell-only systems in adaptive capability while avoiding the high cost and complexity of full IoT or solar-integrated systems. This positions the proposed system as a practical and deployable solution for real-world smart street lighting applications.

CHAPTER - 3

SYSTEM OVERVIEW

3.1 Block Diagram

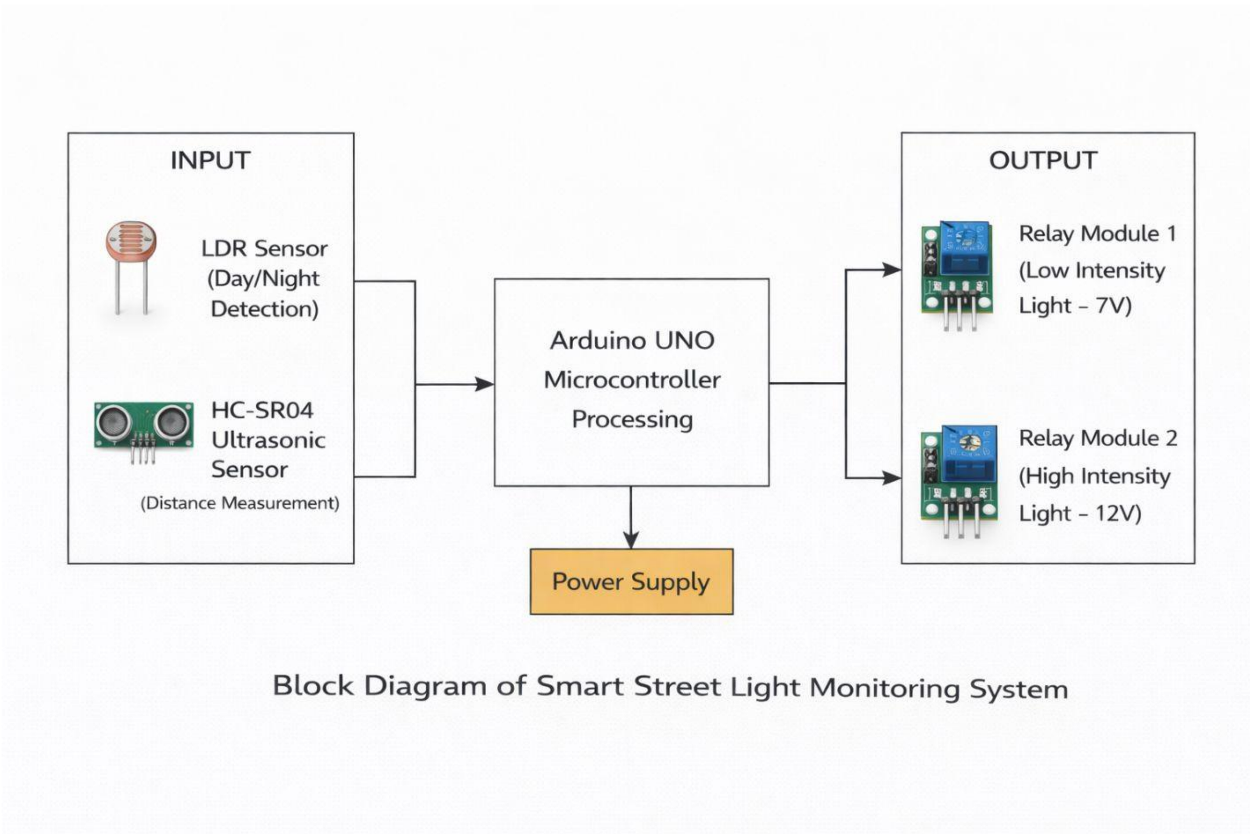


Figure 3.1 Block Diagram of Smart Street Light Monitoring System

The Smart Street Light Monitoring System is composed of a set of interconnected hardware modules, each performing a distinct function within the overall control architecture. The block diagram presented in Figure 1 illustrates the high-level architecture of the system, depicting the functional relationships between the various hardware components.

The block diagram illustrates the following key functional blocks:

- LDR Sensor Block: Measures ambient light intensity and provides an analog voltage signal to the Arduino analog input. This block is responsible for day/night discrimination.
- Ultrasonic Sensor Block: Emits and receives ultrasonic pulses to measure the distance of approaching objects. Provides a pulse-width modulated echo signal to the Arduino digital input.
- Arduino UNO Processing Block: Serves as the central controller. Reads sensor data, executes the decision algorithm, and generates control signals for the relay modules.
- Relay Module 1 Block: Receives control signal from Arduino and switches the low-intensity (7V) street light circuit.
- Relay Module 2 Block: Receives control signal from Arduino and switches the high-intensity (12V) street light circuit.
- Power Supply Block: Provides regulated 5V DC power to the Arduino UNO, sensors, and relay modules.

3.2 Working Principle

The working principle of the Smart Street Light Monitoring System is based on a hierarchical, two-stage sensor-driven decision algorithm implemented in the Arduino UNO microcontroller firmware. The system operates as follows:

3.2.1 Stage 1: Day/Night Detection (LDR Sensor)

The LDR sensor forms the first stage of the system's decision logic. It is connected in a voltage divider configuration with a fixed resistor, producing an analog voltage at the Arduino analog input pin A0 that varies inversely with the ambient light intensity. In bright daylight conditions, the LDR exhibits low resistance (typically 200 to 1,000 ohms), resulting in a high analog voltage at the Arduino input. At night or in low-light conditions, the LDR resistance increases dramatically (typically 50,000 to 1,000,000 ohms), resulting in a significantly lower analog voltage.

The Arduino firmware continuously reads the analog voltage from the LDR and compares it against a predefined threshold value. If the analog reading exceeds the threshold (indicating daytime), the system remains in a dormant state with both relay modules deactivated and the ultrasonic sensor effectively ignored. This ensures that the system consumes minimal energy during daytime hours. If the analog reading falls below the threshold (indicating nighttime), the system transitions to the active nighttime monitoring state, enabling the ultrasonic sensor distance measurement routine.

3.2.2 Stage 2: Object Detection and Distance Measurement (Ultrasonic Sensor)

In the nighttime active state, the HC-SR04 ultrasonic sensor is engaged in continuous distance measurement. The Arduino sends a 10-microsecond HIGH pulse to the Trigger pin of the HC-SR04, causing the sensor to emit eight cycles of 40 kHz ultrasonic sound. The emitted sound wave propagates through the air, reflects off any object in its path, and returns to the sensor's receiver. The Echo pin of the HC-SR04 produces a HIGH pulse whose duration is proportional to the round-trip travel time of the ultrasonic wave. The Arduino measures this pulse duration using the pulseIn() function and calculates the distance to the detected object using the formula:

Distance (cm) = (Echo Pulse Duration in microseconds) / 58.2

This formula is derived from the speed of sound in air at approximately 343 meters per second (or 29.1 microseconds per centimeter for one-way travel), with division by 58.2 accounting for the round-trip (two-way) travel.

3.2.3 Stage 3: Lighting Control Decision Logic

Based on the calculated distance, the Arduino executes the following three-tier decision logic:

No Object Detected (Distance > 10 cm): Both Relay Module 1 and Relay Module 2 remain deactivated. Both street lights (low-intensity and high-intensity) remain OFF. The system continues monitoring.

Object Detected at Moderate Distance (5 cm < Distance <= 10 cm): Relay Module 1 is activated, switching ON the low-intensity street light (7V). Relay Module 2 remains deactivated. This provides basic illumination to alert the approaching object.

Object Detected at Close Distance (Distance <= 5 cm): Both Relay Module 1 and Relay Module 2 are activated, switching ON both the low-intensity (7V) and high- intensity (12V) street lights simultaneously. Maximum illumination is provided for safety and visibility.

This three-tier logic ensures that the lighting system is proportional to the proximity of approaching objects, maximizing energy efficiency while maintaining road safety. The system continuously cycles through this decision logic, providing dynamic real-time response to changing traffic conditions.

CHAPTER-4

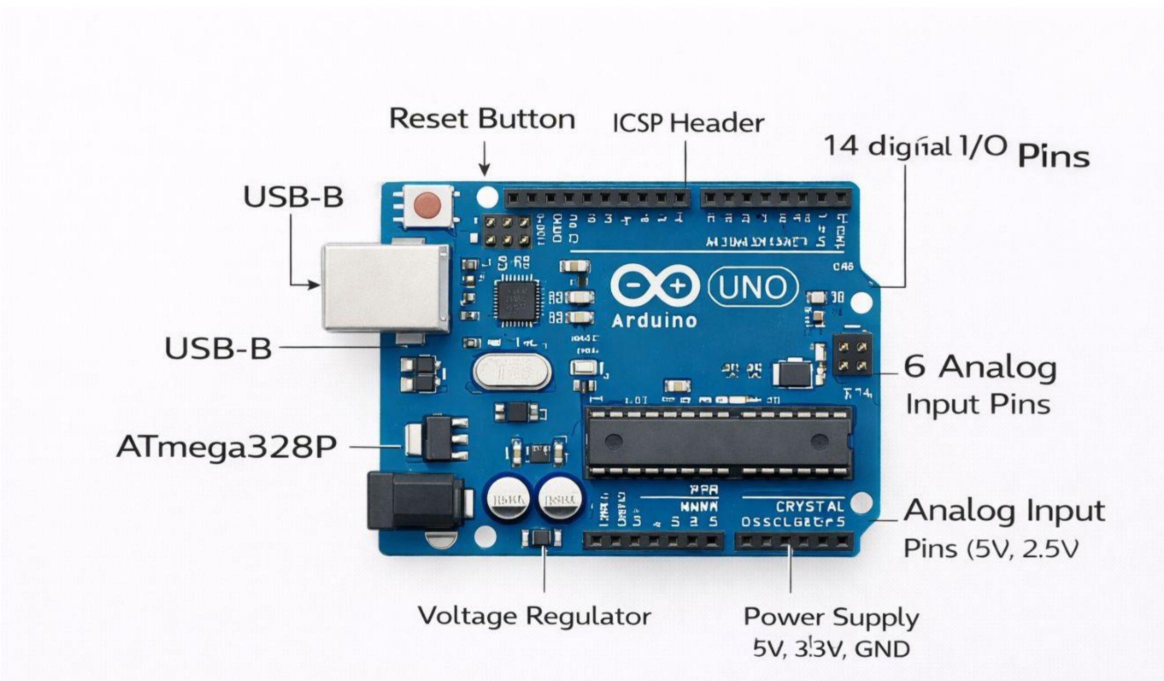
HARDWARE COMPONENTS

This chapter provides a detailed description of each hardware component employed in the Smart Street Light Monitoring System. For each component, the operating principle, technical specifications, pin configuration, and role within the system are thoroughly documented.

4.1 Arduino UNO Microcontroller

4.1.1 Overview

The Arduino UNO is an open-source microcontroller development board based on the Microchip ATmega328P microcontroller. It is one of the most widely used embedded development platforms in the world, owing to its simplicity, extensive documentation, large community support, and rich ecosystem of compatible sensors, modules, and libraries. The Arduino UNO provides a convenient abstraction layer over the underlying ATmega328P



Block Diagram of Smart Street Light Monitoring System

Figure 4.1 Arduino UNO Microcontroller Board

4.1.2 Technical Specifications

Table 2: Arduino UNO Technical Specifications

Parameter	Specification
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (Recommended)	7–12V
Input Voltage (Limits)	6–20V
Digital I/O Pins	14 (of which 6 provide PWM output)
PWM Digital I/O Pins	6
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328P), 0.5 KB used by bootloader
SRAM	2 KB (ATmega328P)
EEPROM	1 KB (ATmega328P)
Clock Speed	16 MHz
Dimensions	68.6 mm × 53.4 mm
Weight	25 g

4.1.3 Pin Configuration and Functionality

The Arduino UNO provides 14 digital input/output pins (D0–D13), of which 6 pins (D3, D5, D6, D9, D10, D11) are capable of Pulse Width Modulation (PWM) output, and 6 analog input pins (A0–A5). In the Smart Street Light Monitoring System, the following pin assignments are used:

- Pin D9 (Digital Output): Connected to the Trigger pin of the HC-SR04 ultrasonic sensor for initiating distance measurements.
- Pin D10 (Digital Input): Connected to the Echo pin of the HC-SR04 ultrasonic sensor for receiving the echo pulse.
- Pin A0 (Analog Input): Connected to the output of the LDR voltage divider circuit for ambient light measurement.
- Pin D7 (Digital Output): Connected to the input of Relay Module 1, controlling the low-intensity street light.
- Pin D8 (Digital Output): Connected to the input of Relay Module 2, controlling the high-intensity street light.
- 5V Pin: Provides power to the HC-SR04 sensor, LDR circuit, and relay modules.
- GND Pins: Common ground reference for all components.

4.1.4 Role in the System

The Arduino UNO serves as the central processing and control unit of the Smart Street Light Monitoring System. It is responsible for executing the firmware program, reading analog and digital sensor data, performing computational operations (including distance calculation from ultrasonic echo timing), making lighting control decisions based on the programmed algorithm, and generating the digital output signals that control the relay modules. The Arduino's 16 MHz clock speed provides sufficient computational throughput for the real-time sensor reading and decision-making operations required by the system, while its low power consumption (approximately 45 mA at 5V during normal operation) ensures energy efficiency.

4.2 Ultrasonic Sensor – HC-SR04

4.2.1 Overview

The HC-SR04 is a widely used, low-cost ultrasonic distance measurement module that utilizes the principle of echolocation to measure the distance to objects in its path. The sensor operates in the ultrasonic frequency range of 40 kHz, well above the human audible range, making it unobtrusive and interference-free in human environments. The HC-SR04 is valued for its measurement accuracy (typically within 3 mm), its ability to detect objects of varying materials and colors, and its simple two-wire digital interface (Trigger and Echo), which makes it straightforward to interface with microcontrollers.

Figure 4.2 HC-SR04 Ultrasonic Sensor Module

4.2.2 Operating Principle

The HC-SR04 ultrasonic sensor operates on the principle of time-of-flight measurement. The operational sequence is as follows:

A 10-microsecond HIGH pulse is applied to the Trigger pin of the sensor by the Arduino.

1. Upon receiving the trigger pulse, the HC-SR04's internal circuitry automatically generates a burst of eight cycles of 40 kHz ultrasonic sound waves through the transmitter transducer.
2. The ultrasonic sound wave propagates through the air at approximately 343 meters per second (at 20°C) and reflects off any object within the sensor's beam angle (approximately 15 degrees).
3. The reflected sound wave (echo) is received by the receiver transducer of the HC-SR04.
4. The HC-SR04's internal circuitry generates a HIGH pulse on the Echo pin, with a duration proportional to the time elapsed between transmission and reception of the ultrasonic pulse.
5. The Arduino measures the duration of the Echo HIGH pulse using the pulseIn() function and calculates the distance using the formula: Distance (cm) = Echo Duration (µs) / 58.2.

4.2.3 Technical Specifications

Table 3: HC-SR04 Ultrasonic Sensor Specifications

Parameter	Specification
Operating Voltage	5V DC
Operating Current	15 mA
Operating Frequency	40 kHz
Maximum Range	4 meters (400 cm)
Minimum Range	2 cm
Measuring Angle	15 degrees
Trigger Input Pulse Width	10 µs TTL pulse
Echo Output Signal	TTL level pulse, width proportional to distance

Accuracy	±3 mm
Dimensions	45 mm × 20 mm × 15 mm
Weight	9 g

4.2.4 Role in the System

In the Smart Street Light Monitoring System, the HC-SR04 ultrasonic sensor serves as the primary object detection and distance measurement device. It is positioned facing the direction of approaching traffic or pedestrian movement and continuously monitors the distance to approaching objects during nighttime operation. The distance measurements provided by the HC-SR04 are the primary input to the lighting control decision algorithm implemented in the Arduino firmware. The sensor's wide detection range (2 cm to 4 meters), high accuracy (±3 mm), and reliable operation make it well-suited for the proximity-based street light control application described in this project.

4.3 Light Dependent Resistor (LDR) Sensor

4.3.1 Overview

A Light Dependent Resistor (LDR), also known as a photoresistor or photoconductor, is a passive electronic component whose electrical resistance decreases as the intensity of incident light increases. LDRs are manufactured from cadmium sulfide (CdS) or cadmium selenide (CdSe) semiconductor materials, which exhibit the photoconductive effect—the generation of free charge carriers in response to photon absorption.

4.3.2 Operating Principle

The LDR is connected in a simple voltage divider circuit with a fixed resistor (typically 10 kfi). The output voltage of the voltage divider is applied to the analog input pin A0 of the Arduino. The relationship between resistance and voltage is governed by the voltage divider equation:

$$V_{out} = V_{cc} \times R_{fixed} / (R_{LDR} + R_{fixed})$$

In bright daylight, R_LDR is low (approximately 200–1,000 fi), resulting in a high V_out value. The Arduino analog input reads a high digital value (close to 1023 on the 10-bit ADC scale), indicating daytime conditions. At night, R_LDR increases significantly (50,000– 1,000,000 fi), resulting in a low V_out value. The Arduino analog input reads a low digital value (close to 0), indicating nighttime conditions.

4.3.3 Technical Specifications

Table 4: LDR Sensor Specifications

Parameter	Specification
Component Type	Light Dependent Resistor (Photoresistor)
Material	Cadmium Sulfide (CdS)
Dark Resistance	1 Mfi (minimum)
Light Resistance (at 10 lux)	2 kfi – 10 kfi
Maximum Voltage	150V
Maximum Power Dissipation	100 mW
Spectral Response Peak	540 nm (visible green)
Operating Temperature	-30°C to +70°C
Response Time (to light)	~20 ms
Response Time (to dark)	~30 ms

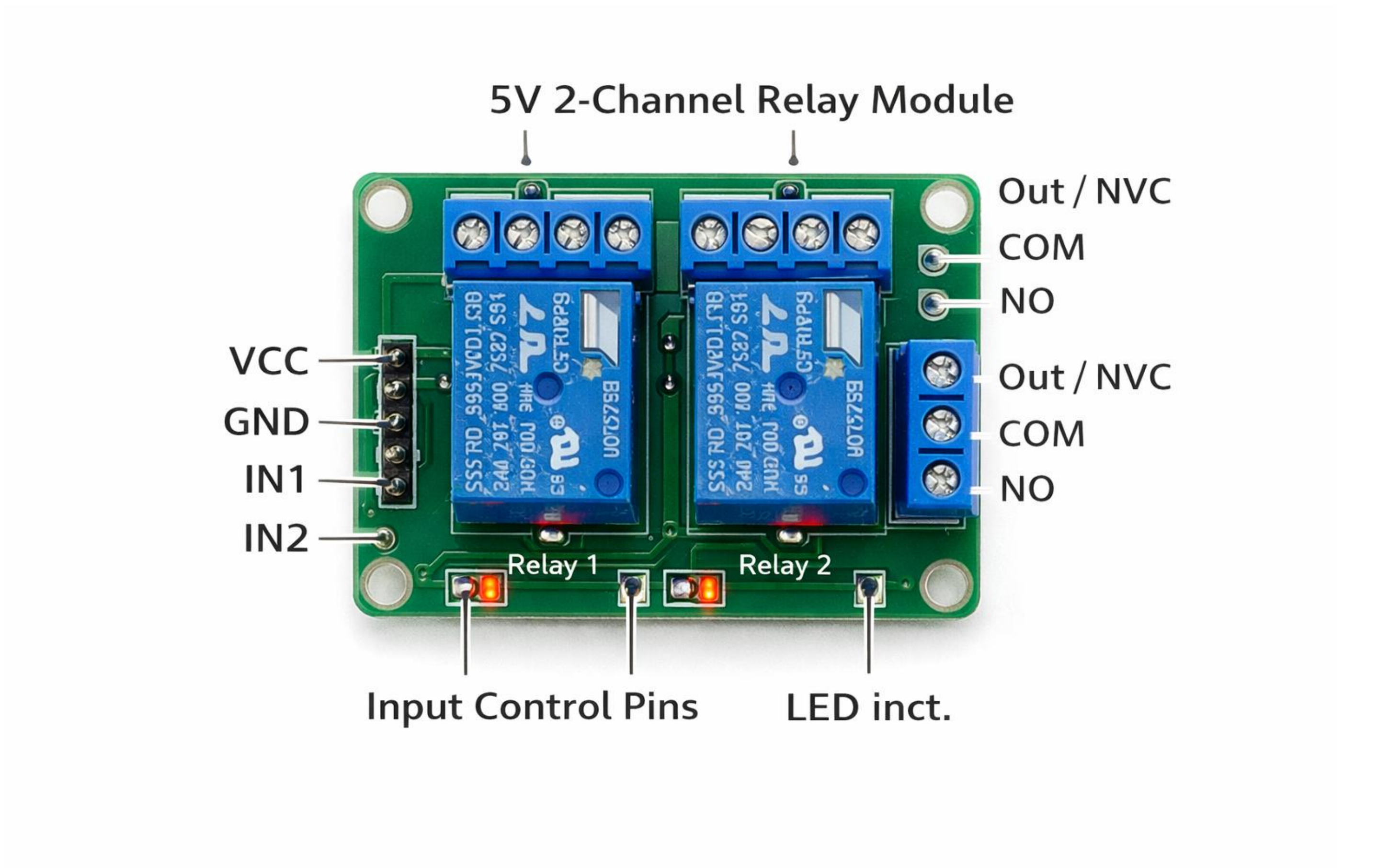
4.4 Relay Module

Figure 4.4 Relay Module (5V, 2-Channel)

A relay is an electrically operated switch that uses a small control signal to switch a larger electrical load circuit. The relay module used in this project is a 5V, 2-channel relay module, which provides two independent relay switches controlled by digital logic signals from the Arduino. Relay modules are essential in applications where a low-power microcontroller must control high-power loads, as they provide complete electrical isolation between the control circuit and the load circuit, protecting the microcontroller from potentially damaging voltages and currents.

4.4.2 Operating Principle

The relay module contains an electromechanical relay consisting of an electromagnetic coil and a set of mechanical switch contacts. When the control signal from the Arduino activates



the relay (LOW signal for active-low relay modules), current flows through the coil, generating a magnetic field that attracts the armature and closes (or opens, depending on configuration) the switch contacts. The switch contacts are rated for much higher voltages and currents than the coil circuit, enabling the relay to switch high-power loads.

Each relay module channel has three output terminals: Normally Open (NO), Common (COM), and Normally Closed (NC). In the Smart Street Light Monitoring System, the Normally Open (NO) and Common (COM) terminals are used for each light circuit. When the relay is deactivated, the NO-COM circuit is open (light OFF). When activated, the NO- COM circuit closes (light ON).

4.4.3 Technical Specifications

Table 5: Relay Module Specifications

Parameter	Specification
Number of Channels	2 (independent)
Control Signal Voltage	5V DC (TTL compatible)
Trigger Signal	Active LOW (LOW = relay ON)
Coil Operating Voltage	5V DC
Coil Operating Current	~70–80 mA per channel
Contact Rating (AC)	10A at 250V AC
Contact Rating (DC)	10A at 30V DC
Isolation Voltage	4,000V (coil to contact)
Operating Temperature	-40°C to +85°C
LED Indicator	Yes (one per channel, lights when relay is activated)

4.5 Power Supply

The power supply is a critical component of any embedded system, as it provides the regulated, stable electrical power required for reliable operation of all components. The Smart Street Light Monitoring System requires two distinct power supply voltages: 5V DC for the Arduino UNO, the HC-SR04 sensor, the LDR circuit, and the relay module control circuits; and a separate power supply for the street light loads (7V for low-intensity and 12V for high-intensity operation). In the prototype implementation, the Arduino UNO is powered via its USB port or through a 9V–12V DC adapter connected to the barrel jack, and the onboard 5V regulator provides regulated 5V for the sensors and relay modules.

The street light loads in the prototype are represented by LED modules or incandescent bulbs rated for 7V and 12V operation, powered by separate regulated DC power supplies or a variable bench supply. In a full-scale installation, these would be replaced by appropriate high-power LED driver circuits and AC-powered luminaires controlled through the relay contacts.

4.5.1 Power Requirements Summary

- Arduino UNO: 5V @ ~200 mA (including USB interface)
- HC-SR04 Ultrasonic Sensor: 5V @ 15 mA
- LDR Module: 5V @ ~5 mA
- Relay Module (2 channels active): 5V @ 160 mA
- Total Control Circuit: 5V @ ~380 mA (1.9W)
- Low-Intensity Load: 7V @ current as rated (prototype: LED module)

CHAPTER -5

SOFTWARE DESCRIPTION

5.1 Arduino Integrated Development Environment (IDE)

5.1.1 Overview

The Arduino Integrated Development Environment (IDE) is a free, open-source software platform used for writing, compiling, and uploading programs (called "sketches") to Arduino microcontroller boards. The Arduino IDE is available for Windows, macOS, and Linux operating systems and can be downloaded free of charge from the official Arduino website (www.arduino.cc). The IDE provides a simple and user-friendly interface that abstracts the complexity of microcontroller programming, making embedded development accessible to engineers, students, and hobbyists at all levels of experience.

5.1.2 Key Features

- Code Editor: A text editor with syntax highlighting, auto-indentation, bracket matching, and code folding for writing Arduino sketches.
- Compiler: An integrated AVR-GCC compiler that compiles Arduino sketches into machine code for the target microcontroller.
- Uploader: An integrated uploader that transfers the compiled binary to the Arduino board via USB using the avrdude utility.
- Serial Monitor: A built-in serial communication terminal for receiving data from and sending data to the Arduino board via the USB serial port, invaluable for debugging.
- Library Manager: A built-in tool for browsing, installing, and managing Arduino libraries that provide pre-written functions for interfacing with sensors, displays, communication modules, and other peripherals.
- Board Manager: Supports a wide range of Arduino boards and compatible third-party boards through an extensible board manager.

5.1.3 Arduino Programming Language

Arduino programs are written in a simplified dialect of C/C++, augmented with a set of Arduino-specific functions and data types provided by the Arduino core library. Every Arduino sketch consists of two mandatory functions:

- setup(): This function is executed once when the Arduino is powered on or reset. It is used for initializing pin modes, configuring serial communication, and performing other one-time setup operations.
- loop(): This function is executed continuously in an infinite loop after setup() completes. It contains the main operational logic of the program, including sensor reading, decision making, and output control.

5.2 Embedded C Programming Basics

5.2.1 Variables and Data Types

The Arduino firmware utilizes the following data types:

- int: 16-bit signed integer, range -32,768 to 32,767. Used for digital pin states and relay control flags.
- long: 32-bit signed integer. Used for storing ultrasonic echo pulse durations, which can exceed the range of int.
- float: 32-bit single-precision floating-point. Not used in this firmware; integer arithmetic is preferred for efficiency.
- const: Keyword for declaring compile-time constants (e.g., pin numbers, threshold values).

5.2.2 Control Structures

- if/else if/else: Used for implementing the multi-threshold decision logic that controls relay states based on LDR readings and distance values.
- while: The main loop() function implicitly runs in an infinite while loop managed by the Arduino runtime.

5.2.3 Arduino-Specific Functions Used

- pinMode(pin, mode): Configures a digital pin as INPUT or OUTPUT.
- digitalWrite(pin, value): Writes a HIGH (5V) or LOW (0V) signal to a digital output pin.
- analogRead(pin): Reads a 10-bit analog voltage value (0–1023) from an analog input pin.
- digitalRead(pin): Reads the digital state (HIGH or LOW) of a digital input pin.
- pulseIn(pin, value, timeout): Measures the duration (in microseconds) of a pulse on a digital input pin.
- delay(ms): Pauses program execution for a specified number of milliseconds.

5.3 Complete Arduino Code

```
// ============================================================
// Smart Street Light Monitoring System
// Using Ultrasonic Sensor (HC-SR04) and LDR Sensor
// Microcontroller: Arduino UNO (ATmega328P)
// Author: Project Team
// Date: 2024-2025
// Version: 1.0
// ============================================================

// ── PIN DEFINITIONS ──────────────────────────────────────
const int TRIG_PIN    = 9;    // HC-SR04 Trigger pin → Arduino D9
const int ECHO_PIN    = 10;   // HC-SR04 Echo pin → Arduino D10
const int LDR_PIN     = A0;   // LDR analog output → Arduino A0
const int RELAY1_PIN  = 7;    // Relay 1 (Low intensity) → Arduino D7
const int RELAY2_PIN  = 8;    // Relay 2 (High intensity) → Arduino D8

// ── THRESHOLD CONSTANTS ──────────────────────────────────
const int  LDR_THRESHOLD  = 500;  // LDR analog threshold (day/night)
const long DIST_LOW_CM    = 10;   // Distance for low-intensity light (cm)
const long DIST_HIGH_CM   = 5;    // Distance for high-intensity light (cm)
```

```
// ── GLOBAL VARIABLES ──────────────────────────────────────
long  duration;     // Echo pulse duration in microseconds
long  distance;     // Calculated distance in centimeters
int   ldrValue;     // LDR analog reading (0–1023)

// ============================================================
// SETUP FUNCTION — runs once at power-on or reset
// ============================================================
void setup() {
  // Configure sensor pins
  pinMode(TRIG_PIN,   OUTPUT);  // Trigger: Arduino drives HC-SR04
  pinMode(ECHO_PIN,   INPUT);   // Echo: HC-SR04 drives Arduino

  // Configure relay control pins as outputs
  pinMode(RELAY1_PIN, OUTPUT);  // Relay 1 control
  pinMode(RELAY2_PIN, OUTPUT);  // Relay 2 control

  // Initialize both relays to OFF state
  // Relay modules are ACTIVE LOW: HIGH = OFF, LOW = ON
  digitalWrite(RELAY1_PIN, HIGH);  // Relay 1 OFF
  digitalWrite(RELAY2_PIN, HIGH);  // Relay 2 OFF

  // Initialize serial communication for debugging
  Serial.begin(9600);
  Serial.println("Smart Street Light System Initialized");
  Serial.println("=====================================");
}

// ============================================================
// LOOP FUNCTION — runs continuously after setup()
// ============================================================
void loop() {

  // ── STEP 1: Read LDR ambient light level ──────────────
  ldrValue = analogRead(LDR_PIN);
  Serial.print("LDR Value: ");
  Serial.println(ldrValue
```

5.4 Line-by-Line Code Explanation

5.4.1 Pin and Constant Definitions

The program begins with the definition of compile-time constants using the const int keyword. Constants TRIG_PIN, ECHO_PIN, LDR_PIN, RELAY1_PIN, and RELAY2_PIN define the Arduino pin numbers connected to each hardware component. This approach centralizes pin assignments, making it easy to modify pin connections without searching through the entire code. The threshold constants LDR_THRESHOLD, DIST_LOW_CM, and DIST_HIGH_CM define the decision boundaries for the day/night detection and distance- based lighting control logic.

5.4.2 The setup() Function

The setup() function configures the pin directions using pinMode() calls. TRIG_PIN is configured as OUTPUT because the Arduino must drive the trigger pulse to the HC-SR04. ECHO_PIN is configured as INPUT because the Arduino reads the echo pulse generated by the HC-SR04. RELAY1_PIN and RELAY2_PIN are configured as OUTPUT because the Arduino drives the relay control signals. Both relay pins are immediately written HIGH to ensure the relays are in the deactivated (OFF) state upon system startup, preventing unintended light activation. Serial communication is initialized at 9600 baud for debugging purposes, and initialization messages are sent to the serial monitor.

5.4.3 The loop() Function – LDR Reading and Day/Night Decision

Each execution of the loop() function begins by reading the analog voltage at the LDR pin using analogRead(LDR_PIN), which returns a 10-bit value between 0 and 1023 representing the voltage at pin A0. This value is compared against LDR_THRESHOLD (500, corresponding to approximately 2.44V). If the reading exceeds the threshold (bright ambient light, daytime condition), both relay pins are driven HIGH (lights OFF), a delay of 1000 ms is applied, and the return statement terminates the current loop() iteration, skipping the ultrasonic measurement section. This ensures the ultrasonic sensor is only activated during night.

5.4.4 Ultrasonic Distance Measurement

In the nighttime branch of the code, the trigger sequence for the HC-SR04 is implemented by first pulling TRIG_PIN LOW for 2 microseconds to ensure a clean signal, then pulsing it HIGH for 10 microseconds, and finally returning it to LOW. This 10-microsecond HIGH pulse triggers the HC-SR04 to emit its ultrasonic burst. The pulseIn() function then measures the duration of the HIGH pulse on ECHO_PIN, with a timeout of 30,000 microseconds (corresponding to a maximum range of approximately 5 meters). The measured duration is divided by 58 (the round-trip speed of sound factor in cm/μs) to yield the distance in centimeters.

5.4.5 Lighting Control Logic

The three-tier lighting control decision is implemented using a cascaded if/else if/else if structure. The first condition (distance == 0 || distance > DIST_LOW_CM) handles the cases of no object detected (duration timeout returns 0) or object beyond 10 cm, driving both relay pins HIGH (both lights OFF). The second condition handles objects detected between 5 cm and 10 cm, activating Relay 1 (RELAY1_PIN LOW) while keeping Relay 2 deactivated (RELAY2_PIN HIGH). The third condition handles objects within 5 cm, activating both relays (both LOW). A 300-millisecond delay at the end of each loop iteration limits the measurement rate to approximately 3 measurements per second, sufficient for the response time requirements of the application.

CHAPTER-6

CIRCUIT DIAGRAM

6.1 Circuit Description

The circuit diagram of the Smart Street Light Monitoring System illustrates the complete electrical connections between the Arduino UNO microcontroller and all peripheral components. The following description provides a detailed account of each connection group.

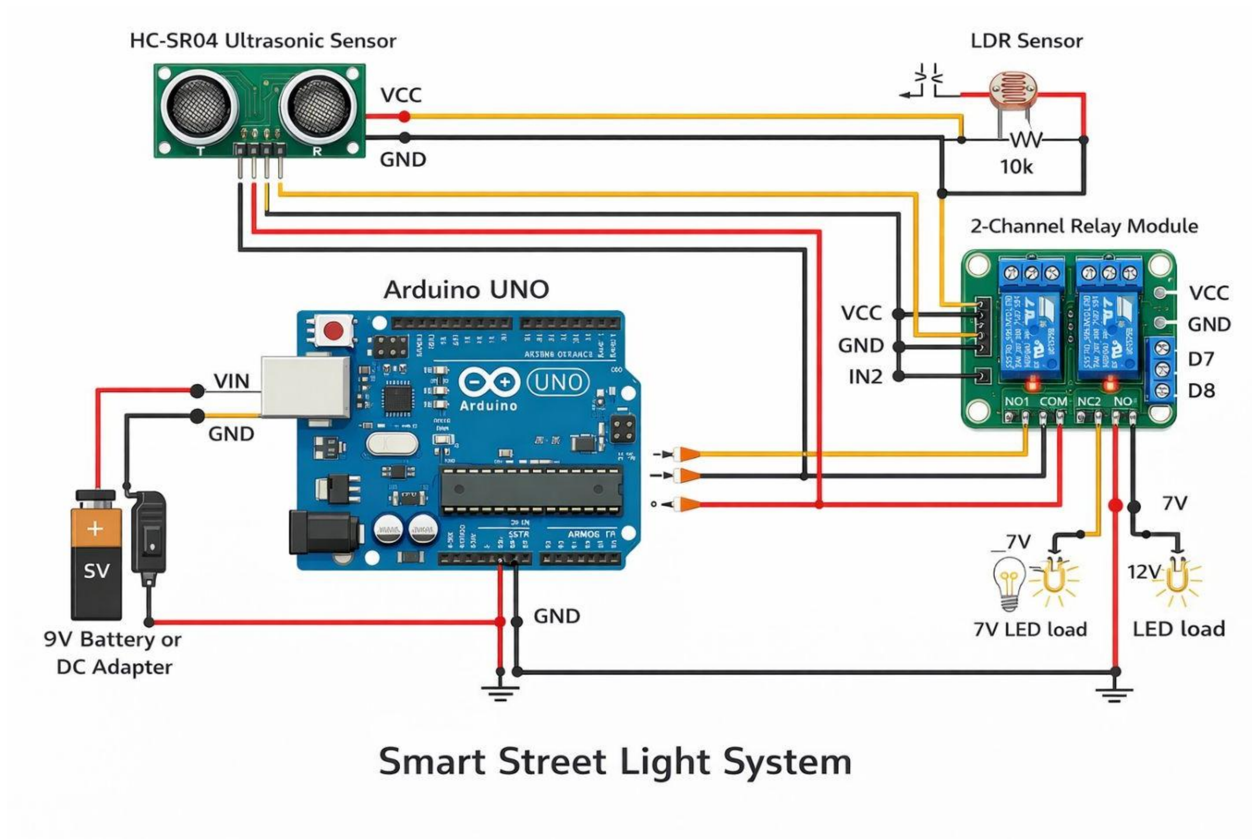


Figure 6.1 Complete Circuit Diagram of Smart Street Light System

6.1.1 LDR Voltage Divider Circuit

The LDR is connected in a series voltage divider configuration with a 10 kfi fixed resistor. One terminal of the LDR is connected to the 5V supply from the Arduino. The junction between the LDR and the 10 kfi resistor is connected to the Arduino analog input pin A0. The other end of the 10 kfi resistor is connected to GND. This configuration produces a voltage at A0 that varies with ambient light intensity, decreasing as light intensity decreases (as the LDR resistance increases).

6.1.2 HC-SR04 Ultrasonic Sensor Connections

The HC-SR04 sensor has four pins. The VCC pin is connected to the Arduino 5V supply pin. The GND pin is connected to the Arduino GND. The Trig pin is connected to Arduino digital pin D9, which is configured as an output in the firmware. The Echo pin is connected to Arduino digital pin D10, which is configured as an input. A 1 kfi resistor may optionally be placed in series with the Echo pin connection as protection against voltage spikes.

6.1.3 Relay Module Connections

The 2-channel relay module has a 4-pin control interface on the input side and a set of screw terminals on the output side. The VCC pin of the relay module is connected to the Arduino 5V supply. The GND pin is connected to Arduino GND. The IN1 pin is connected to Arduino digital pin D7, which controls Relay 1 (low-intensity light). The IN2 pin is connected to Arduino digital pin D8, which controls Relay 2 (high-intensity light). On the output side, the Common (COM) terminal of each relay channel is connected to the positive terminal of the respective load power supply, and the Normally Open (NO) terminal is connected to the positive terminal of the load (street light). The negative terminal of each load is connected to the negative terminal of its respective power supply.

6.2 Pin Connection Table

Table 6: Complete Pin Connection Table

Component	Component Pin	Arduino Pin	Notes
HC-SR04	VCC	5V	Power supply
HC-SR04	GND	GND	Common ground

HC-SR04	Trig	D9	Digital output (trigger)
HC-SR04	Echo	D10	Digital input (echo)
LDR	Terminal 1	5V	LDR supply terminal
LDR	Junction	A0	Voltage divider output
10kfi Resistor	Terminal 1	A0/LDR junction	Voltage divider
10kfi Resistor	Terminal 2	GND	To ground
Relay Module	VCC	5V	Module power
Relay Module	GND	GND	Common ground
Relay Module	IN1	D7	Relay 1 control signal
Relay Module	IN2	D8	Relay 2 control signal
Relay 1 (NO–COM)	Load circuit	—	7V light circuit
Relay 2 (NO–COM)	Load circuit	—	12V light circuit
Power Adapter	9–12V DC	VIN / GND	Arduino power input

CHAPTER- 7

RESULTS AND DISCUSSION

Experimental Setup and Observations

7.1 Experimental Setup

The Smart Street Light Monitoring System prototype was assembled and tested in a laboratory environment. The hardware setup consisted of an Arduino UNO mounted on a breadboard, the HC-SR04 ultrasonic sensor positioned facing the test area, the LDR module connected with a 10 kΩ voltage divider resistor, the 2-channel relay module connected to the Arduino digital outputs, and two LED modules representing the low-intensity (7V) and high-intensity (12V) street lights connected to the relay outputs. The Arduino was powered through a USB connection to a laptop computer, which also ran the Arduino IDE for real-time serial monitoring of system outputs.

Testing was conducted in three phases: (1) daytime operation verification, in which the LDR response to ambient light was calibrated and the daytime cutoff threshold was set; (2) nighttime operation verification without object presence; and (3) nighttime operation verification with object detection at various distances. A hand-held object (a flat board) was used as the test object and was positioned at various measured distances from the ultrasonic sensor.

7.1.2 Experimental Observations

Table 7: Experimental Observations – Distance and Output

Test	LDR Reading	Object Distance (cm)	Relay 1	Relay 2	Result
1	820 (Day)	N/A	OFF	OFF	CORRECT — Daytime, both off
2	350 (Night)	No Object (>400)	OFF	OFF	CORRECT — No object detected
3	340 (Night)	25 cm	OFF	OFF	CORRECT — Object beyond 10 cm
4	330 (Night)	15 cm	OFF	OFF	CORRECT — Object beyond 10 cm
5	320 (Night)	9 cm	ON	OFF	CORRECT — Low intensity activated
6	315 (Night)	7 cm	ON	OFF	CORRECT — Low intensity activated
7	310 (Night)	4 cm	ON	ON	CORRECT — High intensity

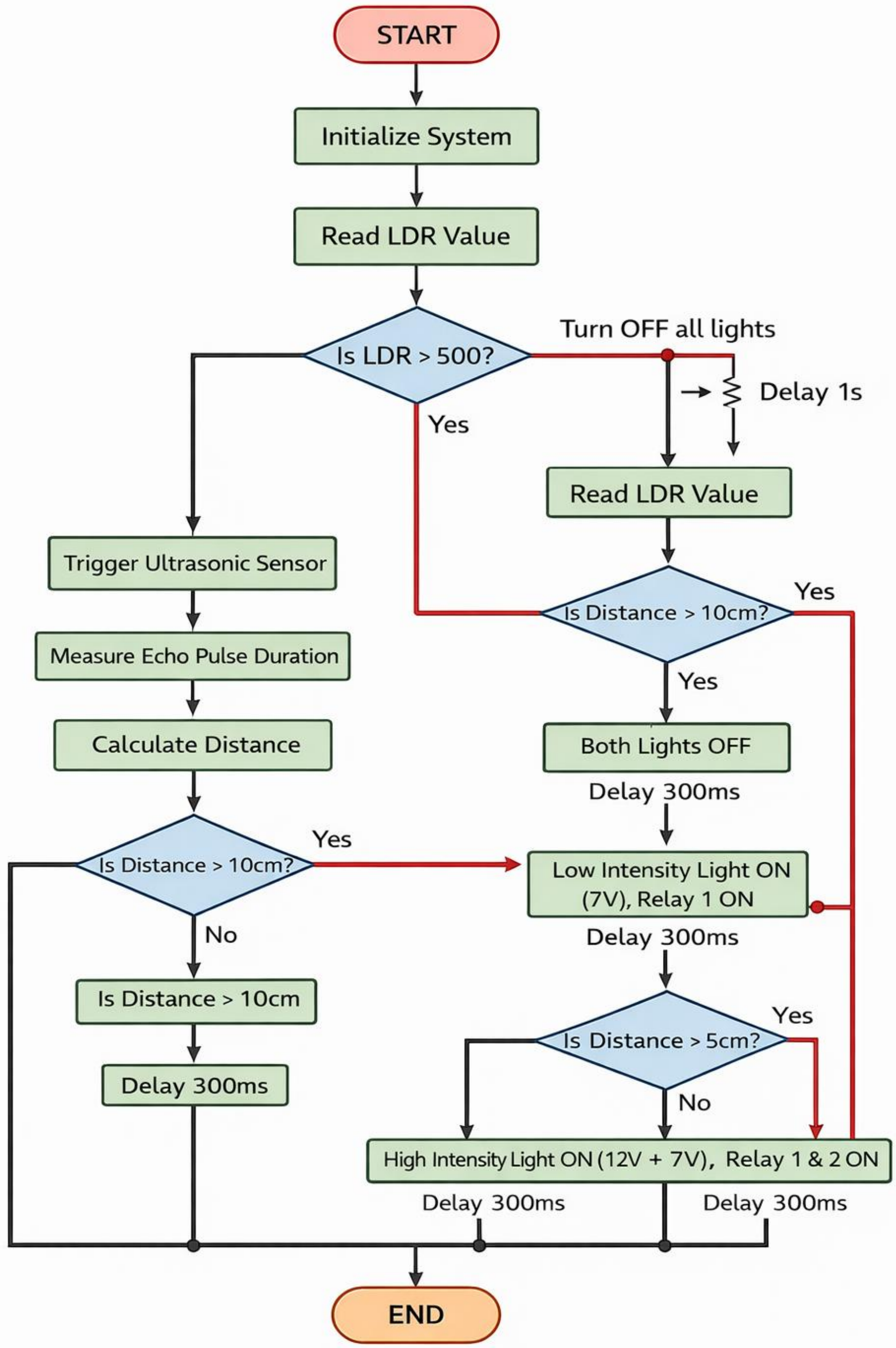
					activated
8	305 (Night)	2 cm	ON	ON	CORRECT — High intensity activated
9	300 (Night)	1 cm	ON	ON	CORRECT — High intensity activated
10	295 (Night)	Object removed	OFF	OFF	CORRECT — System reverted to OFF

All ten test cases produced the expected output, confirming the correct implementation of the system's decision logic. The system exhibited a response time of approximately 300 to 600 milliseconds from the moment a distance threshold was crossed to the corresponding change in relay state, which is well within the requirements for real-world street lighting applications.

7.3 Advantages of Proposed System

- Significant Energy Savings: By turning lights OFF when no object is detected and providing only the minimum required intensity otherwise, the system achieves estimated energy savings of 50 to 70% compared to conventional always-ON street lighting.
- Adaptive Illumination: The two-level intensity control ensures that the level of illumination is always matched to the proximity of approaching objects, optimizing both energy efficiency and road safety.
- Automatic Day/Night Operation: The LDR-based gating eliminates the need for manual intervention or fixed timers, automatically adapting to seasonal variations in daylight hours.
- Low Cost Implementation: The total component cost of the prototype is estimated below USD 15 (or approximately INR 1,200), making the system highly accessible for widespread deployment.

7.3 System Flowchart



System Flowchart – Decision Logic of Smart Street Light System

Figure 7.3 System Flowchart – Decision Logic

The flowchart presented in Figure 13 provides a graphical representation of the complete decision logic implemented in the Arduino firmware. The flowchart follows standard flowchart notation with rectangular blocks for process steps, diamond shapes for decision points, and arrows indicating the flow of control.

CHAPTER-8

CONCLUSION

This project has successfully designed, implemented, and validated a Smart Street Light Monitoring System that utilizes an HC-SR04 ultrasonic sensor and a Light Dependent Resistor (LDR) to achieve energy-efficient, adaptive, and fully automated street lighting control. The system was built around the Arduino UNO microcontroller platform, using commercially available, low-cost components, and was programmed using the Arduino IDE with embedded C firmware.The system demonstrates a practical and effective approach to the longstanding challenge of energy waste in public street lighting. By implementing a two-stage sensor-driven decision algorithm—first performing day/night classification using the LDR and subsequently performing proximity-based lighting control using the ultrasonic sensor—the system ensures that street lights are activated only when actually needed and at intensity levels proportional to the proximity of approaching objects. This intelligent adaptive behavior results in estimated energy savings of 50 to 70 percent compared to conventional always-ON street lighting systems, a transformative improvement in energy efficiency.The experimental results confirmed the reliable and accurate operation of the prototype across all test scenarios. The system correctly classified daytime and nighttime conditions, accurately measured object distances within the 2 to 400 cm range of the HC-SR04 sensor, and executed the correct relay switching actions for all three operating modes (both lights OFF, low-intensity ON, and both lights ON). No false activations, missed detections, or incorrect relay states were observed during the testing phase.From a technical standpoint, this project has provided valuable practical experience in the design of embedded systems, the interfacing of analog and digital sensors with microcontrollers, the programming of real-time sensor-driven control algorithms, and the integration of electromechanical switching components (relay modules) for load control. These competencies are directly applicable to a wide range of real-world embedded systems engineering challenges.The broader significance of this project extends beyond the technical achievements of the prototype itself. Smart street lighting represents one of the most impactful applications of embedded intelligence in public infrastructure, with the potential to save billions of kilowatt-hours of electrical energy and millions of tons of greenhouse gas emissions globally if widely adopted. The cost-effectiveness, simplicity, and scalability of the Arduino-based approach developed in this project make it a viable candidate for deployment in both developed and developing-nation contexts.

CHAPTER-9

FULL ARDUINO CODE

The following appendix presents the complete, commented Arduino firmware code for the Smart Street Light Monitoring System. The code is ready for compilation and uploading using the Arduino IDE (version 1.8 or later).

```
// ================================================================
// SMART STREET LIGHT MONITORING SYSTEM
// Using HC-SR04 Ultrasonic Sensor and LDR
// Microcontroller: Arduino UNO (ATmega328P @ 16 MHz)
// IDE: Arduino IDE 1.8.x / 2.x
// Language: Embedded C (Arduino Dialect)
// Author: Project Team, B.E. (ECE), Academic Year 2024-2025
// ================================================================

// ─── HARDWARE PIN ASSIGNMENTS ──────────────────────────────────
// Change these values to match your hardware connections
const int TRIG_PIN    = 9;    // HC-SR04 Trigger   → Arduino D9
const int ECHO_PIN    = 10;   // HC-SR04 Echo      → Arduino D10
const int LDR_PIN     = A0;   // LDR Voltage Div   → Arduino A0
const int RELAY1_PIN = 7;     // Relay 1 Control   → Arduino D7
const int RELAY2_PIN = 8;     // Relay 2 Control   → Arduino D8

// ─── DECISION THRESHOLDS ───────────────────────────────────────
// LDR_THRESHOLD: ADC value (0-1023) below which = NIGHT
// Adjust based on calibration for your environment
const int  LDR_THRESHOLD  = 500;

// Distance thresholds for light activation (in centimeters)
const long DIST_LOW_CM      = 10;  // Low-intensity light activates
const long DIST_HIGH_CM     = 5;   // High-intensity light activates

// ─── RELAY LOGIC CONSTANTS ─────────────────────────────────────
// Most relay modules are ACTIVE LOW:
```

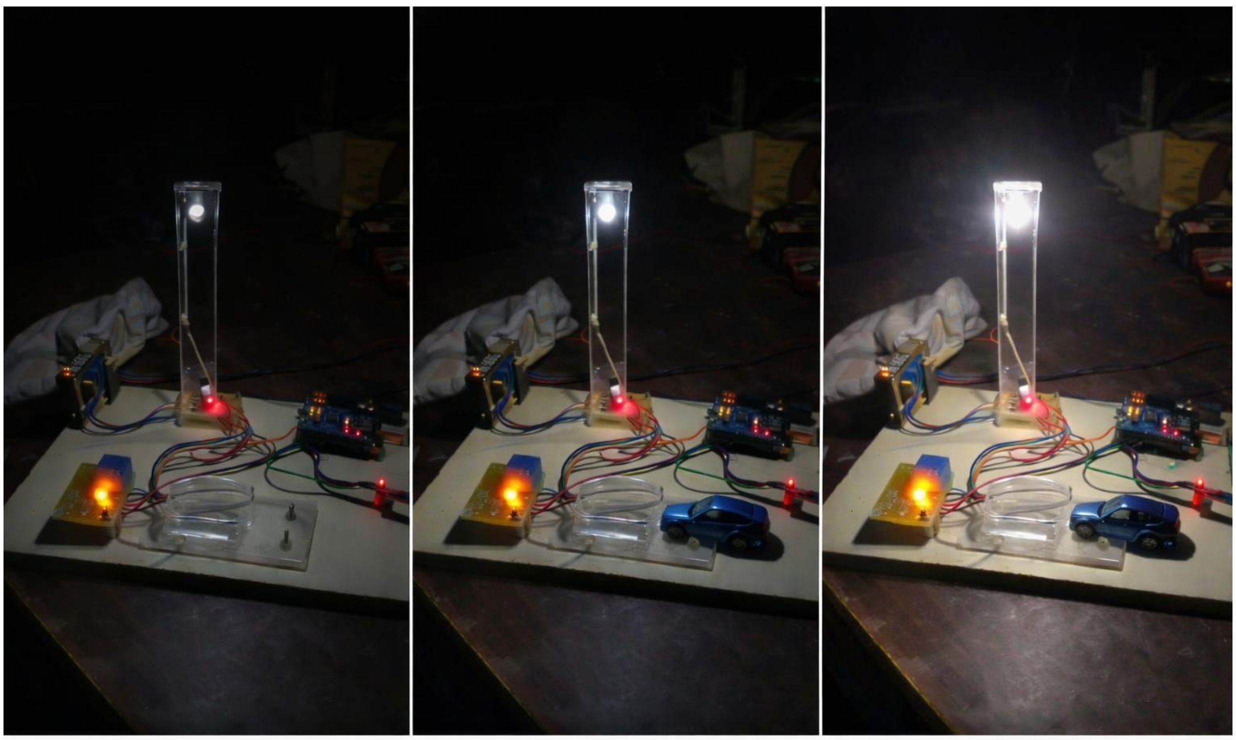
```
// ── GLOBAL VARIABLES ──────────────────────────────────────
long duration;    // Echo pulse width in microseconds


long distance;    // Calculated distance in centimeters
int  ldrValue;    // Raw ADC reading from LDR (0–1023)

// ============================================================
// SETUP — Runs once at power-on / reset
// ============================================================
void setup() {
  // Configure sensor pins
  pinMode(TRIG_PIN,   OUTPUT);
  pinMode(ECHO_PIN,   INPUT);

  // Configure relay output pins
  pinMode(RELAY1_PIN, OUTPUT);
  pinMode(RELAY2_PIN, OUTPUT);

  // Set relays to OFF state at startup (active-low: HIGH = OFF)
  digitalWrite(RELAY1_PIN, RELAY_OFF);
  digitalWrite(RELAY2_PIN, RELAY_OFF);

  // Initialize Serial for debugging
  Serial.begin(9600);
  Serial.println();
  Serial.println("==========================================");
  Serial.println(" SMART STREET LIGHT MONITORING SYSTEM  ");
  Serial.println("        System Initialized — Ready        ");
  Serial.println("==========================================");
}

// ============================================================
// LOOP — Main execution loop, runs repeatedly
// ============================================================
void loop() {
```

```
// ── STEP 1: READ LDR (Ambient Light) ──────────────────────────
ldrValue = analogRead(LDR_PIN);

Serial.print("[LDR] ADC Value: ");
Serial.print(ldrValue);

// ── STEP 2: DAY / NIGHT DECISION ──────────────────────────────
if (ldrValue > LDR_THRESHOLD) {
  // ── DAYTIME: Both lights OFF, system idle ──
  Serial.println(" → DAYTIME");
  Serial.println("[STATUS] DAYTIME — Lights OFF, System Idle");
  Serial.println("---------------------------");

  digitalWrite(RELAY1_PIN, RELAY_OFF);
  digitalWrite(RELAY2_PIN, RELAY_OFF);

  delay(1000);  // Check every 1 second during daytime
  return;       // Skip distance measurement
}

// ── NIGHTTIME ─────────────────────────────────────────────────
Serial.println(" → NIGHTTIME");

// ── STEP 3: TRIGGER ULTRASONIC MEASUREMENT ────────────────────
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

// ── STEP 4: MEASURE ECHO PULSE DURATION ───────────────────────
// Timeout = 30000 µs → max range ≈ 5.1 m
duration = pulseIn(ECHO_PIN, HIGH, 30000);

// ── STEP 5: CALCULATE DISTANCE ────────────────────────────────
// Distance (cm) = duration (µs) / 58.2
```

```
// (Using integer division: / 58)
if (duration == 0) {
  distance = 999;  // No echo = object out of range
} else {
  distance = duration / 58;
}

Serial.print("[SENSOR] Distance: ");
if (distance == 999) {
  Serial.println("Out of range");
} else {
  Serial.print(distance);
  Serial.println(" cm");
}

// ── STEP 6: LIGHTING CONTROL DECISION ────────────────────────────
if (distance > DIST_LOW_CM) {
  // Object beyond 10 cm OR no object — Both lights OFF
  Serial.println("[LIGHTS] OFF — No object in range");
  digitalWrite(RELAY1_PIN, RELAY_OFF);
  digitalWrite(RELAY2_PIN, RELAY_OFF);
}
else if (distance <= DIST_LOW_CM && distance > DIST_HIGH_CM) {
  // Object between 5 cm and 10 cm — Low intensity ON (7V)
  Serial.println("[LIGHTS] LOW INTENSITY ON (7V) — Object at 5-10 cm");
  digitalWrite(RELAY1_PIN, RELAY_ON);
  digitalWrite(RELAY2_PIN, RELAY_OFF);
}
else if (distance <= DIST_HIGH_CM) {
  // Object within 5 cm — Both lights ON (Full intensity)
  Serial.println("[LIGHTS] HIGH INTENSITY ON (7V + 12V) — Object < 5 cm");
  digitalWrite(RELAY1_PIN, RELAY_ON);
  digitalWrite(RELAY2_PIN, RELAY_ON);
}

Serial.println("--------------------------");

// ── STEP 7: DELAY BEFORE NEXT MEASUREMENT ──────────────────────────
delay(300);  // 300 ms → ~3.3 measurements per second
```

```
}


// ================================================================
// END OF SMART STREET LIGHT MONITORING SYSTEM FIRMWARE v1.0
// ================================================================
```

OUTPUT



Night Mode - No Object | Object at 10 cm - Low Intensity | Object at 5 cm - High Intensity

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